Indian Institute of Technology Palakkad

Department of Mechanical Engineering

ME 5610: Soft Computing

Test I (18 Feb 2025)

Time: 8.00 - 8.50 AM **Marks: 20**

Instructions :

(i) **The question paper has TWO pages.**

(ii) **Answer all questions. Questions carry marks as stated against them.**

(iii) **Assume any missing data suitably.**

(iv) **Preferably answer in the same sequence as the questions.**

Q1. The annual operating cost U for an electrical line system is given by the following expression

$$U = \frac{21.9 \times 10^7}{V^2 C} + 3.9 \times 10^6 + 1000C$$

where V is the line voltage in kilovolts and C is the line conductance in mhos.

(a) Find the stationary points for the function, and determine V and C to minimize the operating cost. [4]

(b) Show that the second order sufficient condition is satisfied at the minimum point. [3]

Q2. Using the method of Lagrange multipliers, minimize the following function

$$F(\mathbf{x}) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8$$

subject to

$$x_1 + x_2 = 4$$

[5]

Q3. Answer whether the following statements are **True/False**. Also briefly mention the reasons supporting your answer [$1 \times 5 = 5$]

(i) The gradient of a function at a point gives a local direction of maximum decrease in the function.

(ii) A symmetric matrix is positive definite if its eigenvalues are non-negative.

P.T.O

(iii) A point satisfying first-order necessary conditions for an unconstrained function may not be a local minimum point.

(iv) Optimum design points for constrained optimization problems give stationary value to the Lagrange function with respect to design variables.

(v) If the Hessian of an unconstrained function is indefinite at a candidate point, the point may be a local maximum or minimum.

Q4. Write down the KKT conditions (**Do not solve**) for the following optimization problem

$$\textbf{Minimize } f(x_1, x_2) = 4x_1^2 + 9x_2^2 + 6x_2 - 4x_1 + 13$$

$$\textbf{subject to } x_1 - 3x_2 + 3 \leq 0$$

[3]